

SEMESTER PROJECT: Traffic Light Control System

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Traffic Light Control System Report

Project Objective

The goal of this project is to design and simulate a traffic light system for a 4-way intersection using digital logic principles. The system uses Red, Yellow, and Green LEDs to represent 'Stop,' 'Ready,' and 'Go' signals, respectively. The lights at the intersection operate cyclically, allowing each direction to flow for specific adjustable time periods. The design integrates flip-flops, timers, and logic gates to control the sequence of lights.

Components Used

555 Timer (U1): Operates as an astable multivibrator to generate a clock signal.
74LS90 Counter (U2): A decade counter used to divide the clock and generate timing signals for controlling the LEDs.
Logic Gates:
- AND Gates (74LS08)
- OR Gates (74LS32)
- NOT Gates (74LS04)

LEDs: Represent traffic lights for each signal:
- Red LED: Stop
- Yellow LED: Ready
- Green LED: Go

Resistors and Capacitors: To configure the timing in the 555 timer circuit and manage signal levels.
Power Supply (DC): Provides required voltage to the circuit.

Circuit Description

Clock Generation

The 555 timer (U1) operates in astable mode to generate a continuous square wave signal. This acts as the system clock for the entire circuit. The timing of the clock (frequency) is determined by resistors R1, R2, and capacitor C2.

Frequency formula:

$$f = 1.44 / ((R1 + 2*R2) * C2)$$

State Generation

The clock signal from the 555 timer is fed to the 74LS90 decade counter (U2). This counter generates four sequential states (Q0 to Q3), each representing a specific phase of the traffic light cycle. By decoding the counter outputs using logic gates, we control the activation of Red, Yellow, and Green LEDs for each direction.

Logic for Traffic Lights

The logic for controlling each light (Red, Yellow, Green) is implemented using AND, OR, and NOT gates. Specific combinations of counter outputs enable the corresponding LEDs.

Example:
- Red Signal: Active when Q0 and Q1 are HIGH.
- Green Signal: Active when Q2 is HIGH.
- Yellow Signal: Active during the transition between Green and Red (e.g., Q3).

Adjustable Time Periods

The timing of each state (Red, Yellow, Green) can be adjusted by changing the frequency of the clock signal from the 555 timer.

Truth Table

The truth table for the system maps the counter outputs Q0, Q1, Q2, Q3 to the LED states for each direction:

Counter States	Red (R)	Yellow (Y)	Green (G)
$Q0 = 1$	1	0	0
$Q1 = 1$	1	0	0
$Q2 = 1$	0	0	1
$Q3 = 1$	0	1	0

Karnaugh Maps

Red Light Logic

Simplified Equation:
R = ¬Q2 · ¬Q3

Green Light Logic

Simplified Equation:
G = Q2 · ¬Q3

Yellow Light Logic

Simplified Equation:
Y = Q3 · ¬Q2

Simulation Results

Upon simulating the circuit:
1. The Red LED turns on during states Q0 and Q1.
2. The Green LED turns on during Q2.
3. The Yellow LED turns on during Q3, signaling the transition to the next cycle.

Improvements

Add Pedestrian Signals: Include pedestrian crossing lights synchronized with traffic signals.
Real-Time Adjustments: Implement a microcontroller (e.g., Arduino) for dynamic traffic control based on sensor inputs.
Power Efficiency: Use low-power LEDs and optimize the logic circuit to reduce energy consumption.

Conclusion

The project successfully demonstrates a functional 4-way traffic signal control system using digital logic. It provides a foundation for more complex systems, such as adaptive traffic control with real-time data inputs. The simulation validates the design and ensures correct sequencing of traffic signals.

Introduction

Traffic control systems are essential for ensuring smooth vehicular movement and pedestrian safety in modern urban environments. This project focuses on designing a 4-way intersection traffic light control system using digital logic circuits. The design involves using LEDs to represent traffic lights (Red for Stop, Yellow for Ready, and Green for Go) controlled by a combination of timers, counters, and logic gates. By employing ICs such as the 555 Timer, 4017 Decade Counter, and basic logic gates (AND, NOT), the circuit achieves a cyclic pattern of light changes. Each signal is active for a specific adjustable time, providing a comprehensive solution to manage traffic flow efficiently.

ICs Used

The following Integrated Circuits (ICs) are used in the project:

1. **555 Timer**: Operates in astable mode to generate a clock signal, determining the overall timing of the traffic light sequence.
2. **4017 Decade Counter**: Divides the clock signal into sequential outputs (Q0 to Q9), controlling the phases of traffic signals.
3. **Logic Gates**: Implements the control logic for each light. AND and NOT gates are specifically used for decoding counter outputs.
4. **Resistors and Capacitors**: Additional passive components to configure the timing and stabilize the circuit.
5. **Traffic Lights Module**: Represents the actual traffic lights using LEDs for Red, Yellow, and Green signals.

Truth Table

The truth table defines the states of the traffic lights based on the outputs of the 4017 Decade Counter. Each row corresponds to a specific counter state (Q0 to Q9), and the LED states are determined by logic gates. Here is the detailed truth table:

Counter Output (4017)	Red LED	Yellow LED	Green LED
Q0	1	0	0
Q1	1	0	0
Q2	0	0	1
Q3	0	1	0
Q4	1	0	0
Q5-Q9	0	0	0

Explanation:

1. **Red LED (Stop)**:
    - The Red LED is ON (logic 1) during counter outputs **Q0**, **Q1**, and **Q4**.
2. **Yellow LED (Ready)**:
    - The Yellow LED is ON (logic 1) during counter output **Q3**.
3. **Green LED (Go)**:
    - The Green LED is ON (logic 1) during counter output **Q2**

Karnaugh Maps

Karnaugh Maps (K-Maps) are used to simplify the logic equations for controlling each light. Based on the truth table, the following K-Maps are derived:

Red Light (R)

The Red LED is ON during counter states Q0 and Q1. Using the K-Map, the simplified equation is derived as:
R = Q0 + Q1

Green Light (G)

The Green LED is ON during counter state Q2. Using the K-Map, the simplified equation is:
G = Q2

Yellow Light (Y)

The Yellow LED is ON during counter state Q3. Using the K-Map, the simplified equation is:
Y = Q3

Summary

This project demonstrates a systematic approach to designing a traffic light control system using digital logic circuits. The integration of a 555 Timer for clock generation and a 4017 Decade Counter for state management ensures a reliable and cyclic operation of the traffic lights. Simplified logic using AND and NOT gates further enhances the efficiency of the design. By analyzing the truth table and deriving optimized logic equations through K-Maps, the project provides a practical and scalable solution for real-world traffic management. Future improvements could involve integrating sensors for dynamic traffic control.